JMB Lighting — User Manual

JMB one / 4our / 8ight · Firmware 2.x · 2026-08-20 · 4-port shown (1/8-port variants identical unless noted)

1. Front panel

One rotary encoder: **turn** = scroll / adjust, **click** = select, **hold (1.2 s)** = back one level, **keep holding (3 s)** = straight home + lock from any depth. The home screen shows the JMB mark; hold to unlock into the menu. The panel relocks after inactivity (running tests and effects keep going). When no source is decoding, the lock screen reads **No Data** — whatever the CRMX link state — so a linked-but-silent radio can never masquerade as a live source.

Main menu

Item	What it does
Test Mode	Fade / Step RGBW output tests (click cycles, hold exits)
Ports	per-port or All Ports setup, incl. **Presets** — see below
Sources	**the input hub** — wired DMX and CRMX each with live status (the active source + Hz shows inline and in the header); Chain Listen arms here
Health	worst fault inline; Clear Faults + the live universe viewer
Setup	system only: ID, Bluetooth + PIN, curve, encoder, language, protection, cal, reset

Port Settings (per port, or All Ports)

DMX Address and Head Number open the digit editor (turn = digit, click = next, SAVE commits; hold cancels; the footprint hint shows the span). Personality cycles RGB / RGB16 / CCT / CCT16 / **FX10** (see DMX chart). Effect opens the effect picker. PWM Mode cycles **Smooth 8 kHz / Silent 20 kHz / Camera 62.5 kHz** — Camera is the power-up default and is flicker-free for slow-motion capture (see §8). Current and Temp limits step their trip points and push to the port hardware immediately. Clear Fault re-arms a latched port. **Presets** opens the presets sub-menu (below).

All Ports adds two rows: - **Chain: Local / Global** — Local edits this unit. **Global** also passes a baton down the DMX line after you commit (see §4). - **Apply: Same / Seq / Offset** — how a base address fans out: everyone the same; follow-on by each port's real footprint; or fixed 10-channel spacing (re-personality later without re-patching).

Presets (Port Settings → Presets)

Three rows: - **Dimmer** — a live master brightness. On a single port it opens straight on that port; on **All Ports** it walks each port in turn (turn = level, click = next port). The output follows the encoder **live** — no need to confirm to see it. 1 % steps on a slow turn, coarser when you spin. Hold = cancel and restore the levels you started with. - **Colours** — the factory colour / CCT looks. - **Scenes** — your four user looks (U1–U4).

When Port Settings is on **All Ports** *and* **Chain: Global**, the Dimmer row becomes a **Chain Master**: you set one level and, after a **DEPLOY DIMMER?** confirm, it sets every port on this unit *and* every **armed** unit down the DMX line (see §4). Deploy from the **first** controller in the chain to reach the whole rig.

Sources (the input hub)

One screen answers "what's driving the rig": the header names the **active source and its refresh rate**, and each row carries live status — **DMX In** (Hz or *silent*; click = the line analyser), **CRMX** (*no link / NO DATA /* Hz — data-driven, so "linked but nothing arriving" is visible at a glance; click = link/unlink/antenna) and **Chain Listen** (arms this unit to accept ONE Global baton — always Off at power-up, disarms itself after use).

Setup

Controller ID (shows on screens, in the Bluetooth name after reboot, and in the RDM label) · Bluetooth on/off + PIN · Dimmer Curve (Lin / Soft 1.6 / Thtr 2.2) · Reverse Encoder · Language · Thermal Fold · Sensor Cal · Factory Reset (confirmed).

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2. Output priority

Test mode → Identify (RDM/app) → live DMX → Effect → Manual → hold-last. Live DMX always outranks manual and effects; when the desk vanishes, the last DMX look holds until you set something else. Outputs soft-start over ~1 s at power-up. All sources pass the dimmer curve.

"Live DMX" is the wired input *or* the CRMX radio — they merge into one universe, so everything downstream (patch, FX personality, the diagnostics viewer, RDM) treats the radio exactly like a cable. The header on every screen names the winner: **DMX** or **CRMX**.

2a. Firmware update over Bluetooth (BLE OTA)

The controller updates its own firmware straight over Bluetooth from the app — no USB, no cable, no Wi-Fi.

- **In the app** (Setup → *Firmware update*): pick a `firmware.bin`, confirm, and the app streams it to the controller over the Bluetooth link. A progress bar tracks the transfer; outputs hold their last look throughout.
- **The image is checked before it runs** (CRC verified end-to-end) and only then made the next boot.
- **Automatic rollback.** After an update the controller boots the new image on probation: if it doesn't come up cleanly and confirm itself, the controller falls back to the previous working firmware on its own. The app shows a *confirming…* indicator during probation and *firmware confirmed* once the new build has proven itself, so a bad flash can never brick the unit.

> Wi-Fi / network DMX (sACN, Art-Net) and Wi-Fi OTA are not part of the shipping firmware — the control path is deliberately kept off the network for responsiveness. Wired DMX, CRMX and BLE cover input and updates.

3. Effects

Twenty-one built-in dynamic looks per port (Candle, Fire, Fireworks, Water, Siren, Laser, Club, Lightning, Rainbow, Aurora, Sunset, TV Sim, Strobe, Police, Pulse, Party, Traffic, **Fluoro, Paparazzi, Explosion, Clouds**). Set from Port Settings → Effect, the app, or DMX channel 8 of the FX personality. Speed 0–255 (128 = designed tempo) from the app or DMX channel 9. While an effect runs, the app's **RGB sliders become its colour trim** — all at full = the untouched effect; pull one down to take that colour out (Rainbow minus blue, say). The trim is a live tool: it resets to full at power-up. **Effects never start on their own.** When DMX is absent, an effect runs only from the moment you deliberately pick it (panel or app) — a saved effect does not auto-run at power-up, and a desk dropout (cable or CRMX) always **holds the last DMX look** rather than starting one. Pick the effect again after a loss to run it. Siren/Police alternate across ports like a lightbar; Party pulses fully-saturated colour on a 128 BPM bar clock; Fireworks bursts out of true darkness; **Sunset is a one-shot arc** — from golden hour it eases through ember red down to black over ~2 minutes and stays there; re-select it (or dip the port dimmer to 0 and back up) to run another sunset. Traffic plays passing-headlight sweeps with independent timing per port. **The film gags:** Fluoro is a failing fluorescent tube — rapid failed strikes, then it catches and holds with a faint mains buzz; Paparazzi fires irregular hard white pops, independent per port, like a press pit; Explosion is a white-hot flash blooming to an orange fireball with an ember-flicker decay; Clouds drifts daylight slowly dimmer and cooler as cloud crosses the sun, then back.

Rig-wide effect controls (the app's Effects card): **Tap tempo** sets the whole rig's beat-effect speed to the music — tap the app's Tap button in time (or type a BPM, 60–240; 128 is the designed tempo). **Spread** offsets each port in time (0–100 %) so effects sweep across the rig as a wave instead of every port in lockstep. Both persist across reboots.

Capture to scene: the app's Scenes tab saves the current look — **Save current as scene…** (named) or **Capture live** (one tap, auto-named). A scene captures the whole programmer state (colour, or effect + speed), recalls with a crossfade, and rides in the show file. **Up to 100 scenes per phone** (they live in the phone's browser storage, and in any show file you export).

4. Global addressing (chain baton)

To address a whole chain of controllers from one screen:

1. On **every unit that should take part**: Sources → **Chain Listen** → **ARMED**. (Units left Off are untouched — preset rigs are safe.)
2. On unit 1: Port Settings → All Ports → Chain: **Global**, choose Apply mode, open DMX Address, enter the base, SAVE.
3. Confirm the **DEPLOY CHAIN?** warning. Unit 1 addresses itself, then each armed unit down the line takes the follow-on addresses **and the next controller ID**, shows a receipt, disarms, and relays onward.

Head numbers chain the same way. A unit accepts one baton per arming.

The dimmer chains too. With the same arming in place, Port Settings → All Ports → Chain: Global → **Presets** → **Dimmer** gives a single **Chain Master** level. Set it, confirm **DEPLOY DIMMER?**, and every armed unit down the line takes the same brightness. Only a downstream unit hears the baton, so deploy from the first controller in the chain. (Addresses are untouched — this only moves the dimmer.)

Looks, blackout and full chain too. All Ports → Chain: Global → **Presets** → **Colours** → pick a colour/CCT and confirm **DEPLOY LOOK?** to set that look on every armed unit. Choosing **Off** blacks out the whole chain; **White** takes it to full. The app's **Chain broadcast** card has one-tap **Blackout / Full / DMX In** buttons for the same thing. (Dimmer and looks never re-address or renumber — they only change the output.)

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5. RDM

The controller is a full E1.20 responder: discovery, identify (flashes the ports and strobes the main screen **HERE**), device info, start address, personality — the root device carries the rack view (a SET fans out per the Apply mode) and each port is addressable individually as a sub-device. E1.37-1 dimmer curve and PWM frequency are settable from the desk. The controller also publishes **per-port temperature and current as E1.20 sensors** on the root device ("Port 1 Temp", "Port 1 Current", ...), so any RDM console reads fixture health live. The main screen also flashes the moment discovery finds the unit. The powered DMX thru forwards everything downstream; downstream fixtures' RDM responses are relayed back to the desk automatically.

RDM controller (the app's Diag tab). The controller is also an RDM *controller* for everything plugged into its thru line — a full RDM tester built into the rack. **Scan** discovers every downstream RDM fixture and lists it as a card: model, label, UID, start address, footprint, personality and sensor count. Tap a card to edit it in place — **identify** (flash the fixture to find it), **start address**, **personality**, **label**, and live **sensor reads**. Scanning takes over the output line while it runs, so with an upstream desk live the app asks before taking over; the session releases the thru automatically when you stop (or after 3 minutes idle).

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6. The app

https://app.jmblighting.co.uk/ — runs in **Chrome or Edge** on Android or a desktop (on **iPhone**, open it in the **Bluefy** browser, which supplies the Bluetooth that iOS Safari lacks). Choose **Add to Home screen** for a full-screen, offline-capable app. Tap **Connect**, pick your controller from the list, and enter the PIN (default **0123** — change it in Setup). If a controller has just been reflashed, toggle the phone's Bluetooth off and on once first.

Everything the app sends goes **straight to the controller over Bluetooth** — no account, no cloud, no internet needed once it has loaded, and nothing about your rig leaves the phone. The app also **updates itself**: opened with any connection it loads the newest version, then keeps working from its cached copy at a venue. There is nothing to install from a store and no updates to chase.

Control — the working screen

One card per output. Each card holds the port's **DMX address** and **head number**, its **personality** (RGB or tuneable white, 8- or 16-bit, or the 10-channel effect mode), an **effect** picker with a **speed** control, and the level controls: **R / G / B sliders** — which become **Warm / Cool** on a tuneable-white port (amber and iceblue tracks, the unused Blue row hidden) — plus two large **jog wheels** for **Dim** and **FX speed** that are easy to ride by feel during a take. Tap the **label** slot in a card's header to name the port (up to 16 characters; the name also appears on that port's own screen when the panel is locked). **Clear all** returns every port to a clean base — full white, effects off, dimmer down — after a confirmation.

Touching any colour or level control **takes manual control automatically** and stops a running effect on that port, so a fader never feels dead. When a live **DMX, sACN or Art-Net** signal is driving the rig, a **"Live DMX in control"** banner appears — your manual moves here are overridden until that signal stops, which is *why* the faders would otherwise seem to do nothing.

Open **Advanced & protection** on a card for that port's safety and quality settings: **PWM mode** (the dimming carrier frequency — see §8), **current limit**, **temperature trip**, **dimmer curve** (**Linear / Soft / Theatre** — how fader travel maps to brightness), and a **Clear fault & re-enable** button that appears only while the port is latched.

Colour — dial a look, don't push channels

A console-style picker for the selected ports (choose **All** or one port at the top). Its five columns — **Dim, Hue, Sat, Kelvin** and **ΔUV** — are all views of one colour: nudge Hue/Sat and Kelvin follows; nudge Kelvin and Hue/Sat follows; the last column you touch wins. **ΔUV** is an independent **green/magenta tint** trim for matching a practical or pleasing a camera. On a tuneable-white port the picker becomes **Kelvin-only** with a live warm/cool mix readout. Reach for this when you want to *choose* a colour rather than balance raw R/G/B by hand.

Scenes — save and recall looks

Store the whole rig's current look as a **scene** and bring it back later, each with its own **crossfade time** (0 = snap). **Capture live** snapshots whatever the controller is outputting right now. Scenes live on the phone and travel inside the exported show file, so a saved show reopens identically on another phone. Tap a scene's name to rename it.

Ports — the health glance

A read-only status page — the same telemetry the port screens show, gathered in one place: each port's **patch**, **output current** (calibrated to within about 1 % of a bench meter), **temperature**, any **fault**, and its active **source**. Your quick "is everything healthy and drawing what I expect?" check.

Diag — the built-in DMX analyser

This is where the controller doubles as a **test instrument** for the signal on the wire. Each tool answers a specific question:

- **Signal card** — *Is a signal arriving, and is it healthy?* Shows whether a signal is present and from where (**wired DMX / sACN / Art-Net**), its **frame rate** with a rolling 90-second graph, **slot count**, **start code**, **time since the last packet**, alternate-start-code packet count, and **error counts**. Your first check that the desk is actually reaching the rig at a good rate.
- **Universe analyser** — *What is every channel doing?* Opens full-screen and lays out the whole universe as **numbered channel meters** — the channel number, a live level bar and its value — so you can read

which channel is doing what at a glance and scroll the full 512. Switch between **Live**, **Min**, **Max** and **Activity** views; **Start capture** records the highest and lowest level each channel reaches over a take (that feeds the Min/Max views), and **Activity** highlights which channels are moving. Tap any channel for its detail: level, percentage, the **port it patches to**, its captured range, and how many times it has changed.

- **Flicker finder** — *Which channel is causing a flicker?* The intermittent-flicker hunter. **Arm** it and every single channel-value change on the wire is logged with the old value, the new value and how long ago it happened — so a light that flickers once every few minutes, or a desk quietly stepping a level, is **caught and timestamped** instead of leaving you guessing. It keeps logging even if you leave the tab, so an intermittent fault can be pinned to the exact channel and moment it occurred.

- **RDM universe** — *What's downstream, and can I fix it from here?* Discovers and controls every RDM fixture down the line like a dedicated RDM tester: **Scan** to find them, then read and set each fixture's **address**, **personality**, **label**, **Identify** and **sensor** readings (see §5). Scanning takes over the output line while it runs.

- **Port bus** — a health table for the internal link to each output module (RX count, CRC errors, fault state), for diagnosing a port that has stopped responding.

Setup — controller and housekeeping

Controller **ID** and **Bluetooth name** (what the pairing list shows), **apply mode** (how a change made to "All" fans out across the ports), **CRMX wireless** (link / unlink / antenna select — the status reads **green when linked**), **Firmware update over Bluetooth** (§2a), **Flash to find** (strobes this controller's screen for 3 s so you can pick it out of a rack — outputs untouched), **Reset all ports** with an **Undo**, **show-file export / import** (one **.jmb** file carrying the whole rig — addresses, heads, labels, looks and scenes — reloadable on any phone or controller), **change PIN**, and **Disconnect**.

Effects on tuneable-white ports. On a CW/WW personality the effect list only offers the effects that read on a white strip (Candle, Fire, Lightning, TV Sim, Strobe, Pulse, Fluoro, Paparazzi, Explosion); the colour-only effects are hidden so you can't pick one that would do nothing.

Demo mode. *Explore in demo mode* on the connect screen walks the entire app — every tab, the analyser, even an RDM scan that discovers example fixtures — with no controller present. Nothing a demo user touches goes anywhere; it is for learning the app or showing it off before the hardware is in front of you.

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7. Fault handling

Ports protect themselves: over-current (hiccup ×3 then latch), bolted-short µs trip, over-temperature, ground-leak imbalance, comms-loss failsafe. Latched ports show a flashing fault banner on their screen, the fault word appears in the menu and the app, and Clear Faults / Port Reset re-arms.

Thermal foldback (Setup → Thermal Fold, on by default): before a port hits its over-temp trip, it eases the output down over the last 10 °C (to a 30 % floor) so a hot fixture *rides through* instead of hard-latching mid-show. The latch stays as the real-fault backstop.

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8. Camera & slow-motion work

Each port dims with a high-frequency PWM carrier, set per port under **Port Settings → PWM Mode** (also in the app, and over RDM via MODULATION_FREQUENCY):

Mode	Carrier	Use it for
Camera *(default)*	**62.5 kHz**	Any shoot — flicker-free on high-speed cameras
Silent	20 kHz	Quietest drive (no coil whine) when no camera is present
Smooth	8 kHz	Maximum low-end dimming smoothness, eye-only / theatre

All three carry the full ~16-bit dimming resolution (temporal dithering makes up the difference at the higher carriers), so the picture is identically smooth — only the flicker frequency changes.

Setting up for slow-motion / high-speed capture:

- **Leave every port on Camera (62.5 kHz)** — it's the power-up default. At 62.5 kHz the output is flat through exposures short enough for normal high-speed work (comfortably to ~1000 fps at sensible shutter angles).
- **Keep all ports on the same mode** so multiple fixtures match on camera. Use All Ports → PWM Mode, or set it once over RDM, to change the whole rig together.
- **Extreme frame rates** (multi-thousand fps) with very short exposures: shoot a quick test first. *Any* PWM dimmer can show faint banding when the exposure catches only a few carrier cycles — if you see it, widen the shutter angle or drop the frame rate slightly. Higher isn't selectable beyond 62.5 kHz.
- **Turn strobe looks off for clean capture.** The Strobe / Police effects and the FX personality's shutter channel (CH10) *deliberately* chop the output — that's the effect, not carrier flicker. Set the shutter open / effect Off.
- **Fades and colour moves are flicker-free** at the carrier; dial fade times and the dimmer curve as normal — nothing special needed for the camera.

9. Quick reference

- Default PIN: **0123** · default addresses: sequential RGB
- DMX chart: see `docs/DMX_CHART.md`
- Start address caps at 513 − footprint so the whole personality fits the universe (FX 10ch → max 503); it drops automatically if you switch a port to a larger personality
- Chain Listen is always Off after power-up — arming is deliberate
- The app's Undo restores the state captured before the last Reset-all
- Supply: 24 V standard (12 V supported — match the strip voltage). All protections are current- and temperature-based, so they work identically on either rail; at 12 V each port delivers half the wattage for the same current limit